Contents

Analysis 1

Select a section to jump to its first page. Page numbers refer to this PDF.

	PDF page
Limits, completeness, and proof	2
Study reflections and logical writing	5
Uniform convergence	6
The real numbers and sequence criteria	7
Continuity and sequences	9
Real numbers, limits, and topology	10
Suprema and compactness	13
Further notes: logic and set theory	19
Limits and topological ideas	24
The Cantor set and compactness	26

Analysis 1

I learn analysis by testing what a theorem really assumes. Limits, uniform convergence and counterexamples keep changing the way I think.

Analysis ~ limit ~ approximation ~ inequality

喜欢数学分析的一年结束了，遗憾也该结束了–还是不够专注不够认真被外界打扰被绩点主义困扰，什么时候可以静心更沉淀一下呢。为什么会喜欢数分呢，是大二总是太无聊了喜欢去拿题目找老师询问然后聊天吗，是喜欢去用不同的 counter example 看待定理但无奈接受于自己永远不会举 counter example 的事实吗，是喜欢多种方法思考同一道题目然后感叹学问的魅力吗，是在期末考试中 Archimedean Property 使得 1/n 在 uniform continuity 的 delta 内然后用 squeeze Theorem 证明出来的结果吗，是喜欢数列逼近的直观画图的直观在证明前就一眼看出来结果的小骄傲吗，还是感叹于用 recurrence equation 去简单得到 n 维球体积的直观吗，是体会到尽管是这么历史悠久严谨的微分方程也还是只能 approximately 刻画大自然从而欣然接受自己无限渺小的心态吗，是钦佩于也是经常处于迷茫之中但是还坚定不移的走应用数学纯数学路途的老师们的坚毅吗，还是沉浸于数学和生活道理结合起来的老师们口中的小故事呢，比如 Minkowski 得盲肠炎在过世前感叹竟然死在相对论发展的时代，比起这些，所有遗憾似乎都全部消失。

⟵ **Proof idea**
Begin with epsilon-delta or a sequence characterization.

The completion of Real number

From the rational number to the real number

like

the process of the extension from Borel set to the measurable space.

(some set that are null are not measurable (not borel set) —such as choosing one representator of each the class of the equivalance set of r where r in [0,1] —this set is not measurable since if it is null —contradiction (infinitely sumamation of 0 is not equal to [0,1]'s length 1; if it is measure with length greater than 0 then the infinitely many which should not be 1.)

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

Uniform Convergence is the most interesting topic that I have learned in this module! LOVELOVE, I love analysis!

Weierstrass continuous nowhere differentiable function

This example's proof tells that the example of x to be 0 is important

The series construction method of this proof is very intrinsic.

⟵ **Motivation**
What does the definition really control? Try to draw it before proving it.

Final exam review (most are about uniform convergence, which is really, really intereting for me since I have got some exercises from Prof. Andrew Lin and also Dr. Chi-Kwong Fok has helped me a lot in his office hours)

For $x = 0$ or if x is irrational, and $x \in [c, d]$

$$\sup \left| \frac{x}{n} \left(1 + \frac{1}{n} \right) \right| = \max \left\{ \left| \frac{c}{n} \left(1 + \frac{1}{n} \right) \right|, \left| \frac{d}{n} \left(1 + \frac{1}{n} \right) \right| \right\}$$

$$\Rightarrow \limsup_{n\to\infty} \left|\frac{x}{n}\left(1+\frac{x}{n}\right)\right| = 0 \Rightarrow \text{this could not say anything}$$

For $x = \frac{a}{b}, b > 0$ and a, b are integers.

$$h_n(x) = a + \frac{a}{n}\left(1+\frac{1}{b}+\frac{1}{bn}\right) = a + \frac{a}{n} + \frac{x}{n} + \frac{x}{n^2} \text{ —Since } \frac{a}{b} = x$$

$$|h_n(x) - a| = \left|\frac{a}{n} + \frac{x}{n} + \frac{x}{n^2}\right|$$

Take a to be the sequence n

$$\lim_{n\to\infty} |h_n(x) - a_n| = \lim_{n\to\infty} \left|\frac{n}{n} + \frac{x}{n} + \frac{x}{n^2}\right| = 1 \neq 0$$

By the definition of uniform convergence, we have concluded that $h_n(x)$ is not uniformly convergent.

- Next one

We know that

$$\frac{(-1)^n}{\sqrt{n}} \sin\left(1+\frac{x}{n}\right)$$

is defined on A, which is a compact set and $A \subseteq [-M, M], M > 0$.

This series if from Fourier series but wecan also use this method to prove.

Idea: when n is large enough, $\sin\left(1+\frac{x}{n}\right)$ is always positive and also $|\sin x| \leq |x|, \forall x \in \mathbb{R}$

This means that

$$\sin\left(1+\frac{x}{n}\right) > 0$$

Since $\frac{\sin\left(1+\frac{x}{n}\right)}{\sqrt{n}} > 0$ and it is decreasing, by Alternating test, it is pointwise convergent.

For the uniform convergence,

$$\frac{(-1)^n}{\sqrt{n}} \sin\left(1+\frac{x}{n}\right) = \frac{(-1)^n}{\sqrt{n}}\left(\sin 1 \cos\frac{x}{n} + \cos 1 \sin\frac{x}{n}\right)$$

Now divide this into two part:

For $\cos 1 \cdot \frac{(-1)^n}{\sqrt{n}} \sin\frac{x}{n}$

Since $|\sin x| \leq |x| \leq \frac{1}{\sqrt{n}} \forall x \in \mathbb{R}$

$$-\frac{(-1)^n|x|}{\sqrt{n}\cdot n} \leq \frac{(-1)^n}{\sqrt{n}} \sin\frac{x}{n} \leq \frac{(-1)^n|x|}{\sqrt{n}\cdot n}$$

$$\left|\frac{(-1)^n}{\sqrt{n}} \sin\frac{x}{n}\right| \leq \left|\frac{(-1)^n}{\sqrt{n}}\frac{|x|}{n}\right| = \frac{|x|}{n^{3/2}} \leq \frac{M}{n^{3/2}}$$

By P-test, $\sum_{n=1}^{\infty} \frac{M}{n^{3/2}}$ is convergent.

By Weierstrass M-test for series,

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \sin\frac{x}{n} \text{ converges uniformly.}$$

For $\frac{(-1)^n}{\sqrt{n}} \cos\frac{x}{n}$.

$$\frac{(-1)^n}{\sqrt{n}} \cos\frac{x}{n} = \frac{(-1)^n}{\sqrt{n}} \left(1 - 2\sin^2\frac{x}{2n}\right) = \frac{(-1)^n}{\sqrt{n}} - \frac{2(-1)^n}{\sqrt{n}} \sin^2\frac{x}{2n}$$

For $\frac{(-1)^n}{\sqrt{n}}$, since it is not relevant to x, we only need to consider the second part.

For $\frac{(-1)^n}{\sqrt{n}} \sin^2\frac{x}{2n}$

$$\left|\frac{(-1)^n}{\sqrt{n}} \sin^2\frac{x}{2n}\right| \leq \frac{1}{\sqrt{n}} \left|\frac{x}{2n}\right|^2$$

By p-test, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \left|\frac{x}{2n}\right|^2$ is convergent.

By Weierstrass M-test for series,

$\frac{(-1)^n}{\sqrt{n}} \sin^2\frac{x}{2n}$ converges uniformly.

While the dream of becoming a pure mathematician has faded, my pursuit of Mathematical Analysis remains strong, driven by a desire to express my thoughts with precision and elegance. My ultimate aspiration of Analysis is to present ideas that captivate others, compelling them to listen. This is not just about sharing information; it's about conveying logical insights that have been distilled through my own deep engagement with the material, reflecting a genuine and personal comprehension of the subject matter.

If a question is not trivial, we should know where is the difficulty of it

⟵ **Proof idea**
Identify the exact statement before proving it.

Here the difficulty is uncountable

eg. f is a continuous function on [0,1] $\int_0^1 f(x)x^n dx = 0$, prove that f(x)=0 for all x in [0,1].

we have $\int_0^1 f(x)[a_n x^n + ... + ax + a_0]dx = 0$

We have $\int_0^1 |f(x)|^2 dx = 0$ (Weierstrass Approximation Theorem and Since $ |f(x)|^2$ is not negative, Suppose f(x) is not equal to 0, then we could find a point x0 where f(x0) is not equal to 0, then we could find a small interval around x0 where f(x) is not equal to 0, then this interval's integral could not be =0, which is a contradiction (the meaning of "continuity" is that if a point is greater than 0, then there is a neighborhood of this point where the function is greater than 0.))

We have $|f|^2 = 0$ a.e. $x \in [0,1]$

So $f(x) = 0$ a.e. $x \in [0,1]$

uncountable is the difficulty of this question (density argument : The closure of Q is R means that $\forall x \in R\ \exists x_n \in Q$ such that x_n converges to x)

So we use the proof by contradiction:

epislon-delta language is not important, the most important thing is to understand the essence of the proof and the logic behind it. It is better to see everything intuitively.

eg. The sum of continuous functions

- 2 functions
- n functions
- infinite functions —not continuous! –We could not choose the minimum δ to make the sum continuous.

⟵ **My thought**
What does the definition really control? Try to draw it before proving it.

Now the content follows the time, not summary at all..

Understand intuitively (Never memorize anything in math!)

The first sketch motivates a limit argument through a geometric example before introducing formal notation.

The second sketch compares the same intuitive picture with the symbolic steps of the argument.

118 after mid-term

easy to think wrongly

If $f'(x)$ exists, $(f^2)'= 2f(x)f'(x)$ but the counter direction might not be true

- eg. If f^2 is differentialble and $f^2 > 0$, then |f| is differentiable since we could write |f| as some function of f(x) and f^2 is differentiable. So we could use the chain rule to show that |f| is differentiable. eg. $|f| = e^{1/2lnf^2}$ or just $|f| = \sqrt{f^2}$
- eg. If f^2 is differentiable, and $f^2 > 0$ f may not differentiable since f itself may NOT continuous at any $x \in \mathbb{R}$. f(x) is 1 when x is rational and -1 when x is irrational. f(x) is not continuous at any point in $\mathbb{R}$, so f(x) is not differentiable at any point in $\mathbb{R}$.

Integrability

When we consider a continuous functon's integrability on [a,b], we could divide the function into 2 parts—one is good and the other is bad. For the good part, the uniform continuity guarantes the integrability. For the bad part, we could use the property of boundedness to show that the integral is finite using very thin intervals to cover the bad part. So we could use the property of boundedness to show that the integral is finite using very thin intervals to cover the bad part.

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Write logically

Each global expression should just appear once. When it has been introduced, we can regard it as a fixed thing and expand our proof behind logically.

eg

- Suppose f is differentiable in $(c-\delta, c+\delta) \setminus \{c\}$ for some $\delta > 0$ with $\lim_{x\to c} f'(x) = A$ exists, and f is continuous at c. Show that $f'(c)$ exists and $f'(c) = A$ in the following two ways:
  - By applying Lagrange's Mean Value Theorem.
  - By applying L'Hospital's Rule.

! Explain clearly how continuity of f at c is used in your argument in each case.

⟵ **Proof idea**
Identify the exact statement before proving it.

- proof:

  Let ε > 0. Since $lim_{x\to c} f'(x) = A$ there exists δ > 0 such that ∀x ∈ (c - δ', c + δ') {c}, |f'(x) - A| < ε (*) .

  Now, take x ∈ (c - δ', c + δ') {c}.

  We have that f is continuous on the closed interval between x and c and differentiable on the open interval between x and c.

  So by Lagrange's MVT we have ∃ξ between x and c s.t.

  (f(x) - f(c)) / (x - c) = f'(ξ)

  Hence, we have that

  |(f(x) - f(c)) / (x - c) - A| < ε by (*) because |f'(ξ) - A| < ε.

  Hence $lim_{x\to c}(f(x) - f(c))/(x - c) = A$.

  i.e. f'(c) = A

All ways to distinguish whether it is uniformly continuous

⟵ **My thought**
What does the definition really control? Try to draw it before proving it.

please fill this before your final exam!

practice before my mid-term exam (my way to think that of question is like this)

$f(x) = x^4$ is not uniformly continuous on $\mathbb{R}$.

Choose $x_n = \sqrt[4]{n+1}$, $y_n = \sqrt[4]{n}$.

Since $x^{1/4}$ is differentiable on $[n, n+1]$, by Lagrange's MVT,

$$|x_n - y_n| = \frac{1}{4}\xi^{-\frac{3}{4}}(n+1-n), \text{ where } \xi \in (n, n+1)$$

So $|x_n - y_n| = \frac{1}{4}\xi^{-\frac{3}{4}} < \frac{1}{4}n^{-\frac{3}{4}} = \frac{1}{4\sqrt[4]{n^3}}$

So $0 < |x_n - y_n| < \frac{1}{4\sqrt[4]{n^3}}$

So $\lim_{n\to\infty} |x_n - y_n| = 0$ by squeeze theorem

So for $\varepsilon_0 = \frac{1}{2}$, by the definition of limit, $\forall\delta > 0$, $\exists m > 0$ s.t. for $n_0 \in \mathbb{N} > m$, $|x_{n_0} - y_{n_0}| < \delta$ but $|f(x_{n_0}) - f(y_{n_0})| = 1 > \varepsilon_0 = \frac{1}{2}$,

This satisfies the negation of the definition of "uniformly continuous".

Uniform convergence

⟵ **Motivation**
Which quantifier comes first? The order of choices often decides the theorem.

eg of an interesting function

$$f_n(x) = nx^n(1-x^2)^n$$

This function has a pointwise limit of 0 by squeeze theorem and L'Hospital's theorem.

It is easy to check the change of its limit and integral is not the same, i.e. $lim_n \int_0^1 f_n(x)$ is not equal to $\int_0^1 lim_n f_n(x)$, so it is not uniformly convergent.

If we look at the limit of its supremum, we could find that the supremum of this function is ∞ when n is large enough. So we could use the supremum to show that the limit of this function is not uniformly convergent. This is not trivial to understand since its pointwise limit is 0 but the supremum is ∞. That is because when n is very large, there is a very thin but tall peak, which makes the supremum of this function to be ∞.

Or we also firstly draw this picture and find intuitively that its supremum is ∞ when n is large enough. So we could use the supremum to show that the limit of this function is not uniformly convergent.

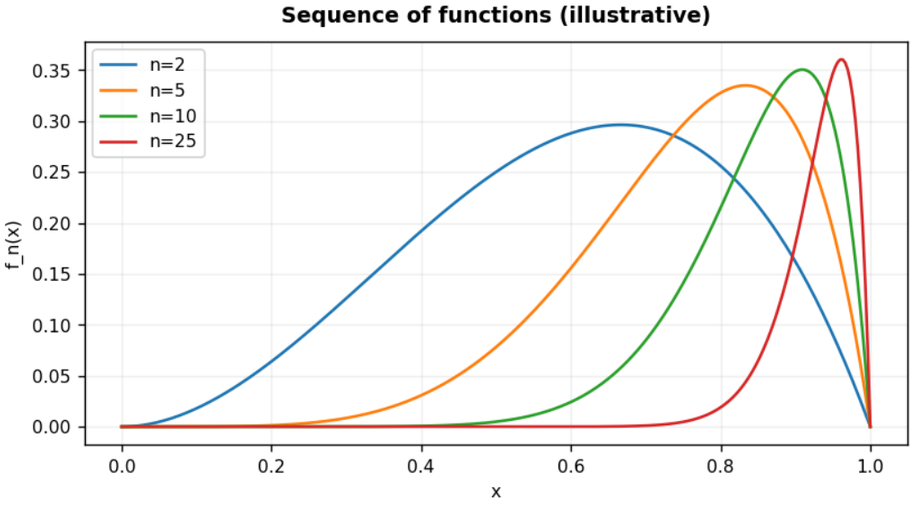


Figure 1: Output from code example 1. Illustrative reconstruction based on nearby code; not the exact original output. R packages: ggplot2; functions: ggplot, geom_line.

Preface

The first section here is the knowledge that attracts most of my interests on this module, followed by the lecture notes I tapped when I am on the journey of this module.

⟵ **Proof idea**
Identify the exact statement before proving it.

The begining main idea of Analysis about the establishment of real number

Motivation

f(x)+f(y)=f(x+y) (addition in linearity in linear transformation)

(a line across the original point)

- 1–>1+1–>1+1+1—>n positive integer(addition is needed/1 is needed)
- 0—>0-1—0-1-1—-n positive integer(addition is needed/0 is needed)
- $\frac{p}{q}$—- rational number(multiplication is needed)

Since the rational number is countable(|N|=|N×N|=|Q|) but irrational number is uncountable, we could not directly go ahead from Q to R\Q. Fortunately, we have the property of density of irrational number(of course also rational number) in R. In this case we could introduce *limit* here, and also **continuity**

- This means that, usually, if a situation in rational number is true, then in the whole real number it is almost true.
- Finally we get a whole line across the original point here.

Example1 If f(x) is continuous on R. g(x)=f(x) when x is a rational number. Then we have g(x)=f(x) on R.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

Proof Use sequence to think intuitively and write down the proof rigorously:

Without loss of generalization, here we let our R-continuous function to be x^2.

question: If f is continuous on [a,b], f(x)=x^2 if x ∈ R\Q. Show: f(x)=$x^2, ∀x∈R $ in [a,b]

proof:

Firstly we write down **Sequential criterion of limit** which we will use to finish the proof:

(Sequential criterion for limits of functions). Let $f : A \to \mathbb{R}$. The following are equivalent. 1. $\lim_{x\to a} f(x) = \ell$. 2. For any sequence $(a_n)_{n\in\mathbb{N}}$ with $a_n \in A, a_n \neq a$ and $\lim_n a_n = a$, we have

$$\lim_n f(a_n) = \ell$$

Proof. (1) $\Rightarrow$ (2): Suppose $\lim_{x\to a} f(x) = \ell$. Then $\forall \varepsilon > 0, \exists \delta > 0$ such that $\forall x$ with $|x - a| < \delta$ and $x \neq a$, we have

$$|f(x) - \ell| < \varepsilon$$

Let $\lim_n a_n = a, a_n \in A$ and $a_n \neq a$ for all $n \in \mathbb{N}$. Then for the $\delta > 0$ above corresponding to $\varepsilon, \exists k \in \mathbb{N}$ such that $\forall n \geq k, n \in \mathbb{N}$, we have

$$|a_n - a| < \delta$$

which implies

$$|f(a_n) - \ell| < \varepsilon.$$

That shows that $\lim_n f(a_n) = \ell$. (2) $\Rightarrow$ (1) : Suppose for the sake of contradiction that $\lim_{x\to a} f(x) \neq \ell$. Then $\exists \varepsilon_0 > 0$ such that $\forall \delta > 0, \exists x$ with $|x - a| < \delta$ and $x \neq a$ such that

$$|f(x) - \ell| \geq \varepsilon_0$$

We construct a sequence as follows. Take $\delta = \frac{1}{n}$. Then $\exists a_n$ with $|a_n - a| < \delta = \frac{1}{n}$ and $a_n \neq a$ such that $|f(a_n) - \ell| \geq \varepsilon_0$.$a_n$ satisfies

$$a - \frac{1}{n} < a_n < a + \frac{1}{n}.$$

By letting $n \to +\infty$ and using the Squeeze Theorem(or we could use another method to finish argument here by constructing only one sequence (choose δ to be $1/n$ also and then always find one x_n in the $\delta-$ neighborhood and this x_n is also convergent to a)), we have $\lim_n a_n = a$. So by the given condition, $\lim_n f(a_n) = \ell$. However, for all $n \in \mathbb{N}$,

$$|f(a_n) - \ell| \geq \varepsilon_0 \Rightarrow f(a_n) \geq \ell + \varepsilon_0 \text{ or } f(a_n) \leq \ell - \varepsilon_0.$$

contradicting to f(a_n) is convergent.

Now we begin our proof:

We have known that if $a_n \in$ R/Q f(a_n)=${a_n}^2$

Since we have the property of “irrational numbers are dense on R” we know that $\exists a_n \in (a - 1/n, a + 1/n), \forall n \in N, a \in Q$. This constructs our sequence a_n such that $lim_{n\to\infty} a_n = a, a \in Q$

This implies that $lim_{n\to\infty} f(a_n) = f(lim_{n\to\infty} a_n)$=f(a).

(The first equality is because of this proposition:(Limit of composition of functions). Let $f : A \to \mathbb{R}, g : B \to \mathbb{R}$ such that $R(f) \subseteq B$. If $\lim_{x\to a} f(x) = \ell$ and $g(x)$ is continuous at ℓ, i.e. $\lim_{x\to\ell} g(x) = g(\ell)$, then

$$\lim_{x\to a} g(f(x)) = g(\ell) = g\left(\lim_{x\to a} f(x)\right).$$

proof:Proof. $\forall \varepsilon > 0, \exists \delta > 0$ such that $\forall y$ with $|y - \ell| < \delta$ and $y \in B$, we have

$$|g(y) - g(\ell)| < \varepsilon$$

For this $\delta, \exists \delta' > 0$ such that $\forall x$ with $0 < |x - a| < \delta'$ and $x \in A$, we have

$$|f(x) - \ell| < \delta$$

So if $0 < |x-a| < \delta'$ and $x \in A$, then $|f(x)-\ell| < \delta$ and $f(x) \in B$, which imply that $|g(f(x))-g(\ell)| < \varepsilon$. This proves the proposition.)

Also, $lim_{n\to\infty} f(a_n) = lim_{n\to\infty} {a_n}^2 = (lim_{n\to\infty} a_n)^2 = a^2$

So we have f(a)=a^2.

example 2

⟵ **Motivation** Why does this definition matter later? Try an example and a counterexample.

This could prove the definiteness in an innner product while also show the good quality of continuous function—-one point>0, the whole neighborhood>0

If a continuous function is positive at one point, nearby values remain positive.

A sufficiently small neighborhood around the point keeps the continuous function above zero.

Continuous function

Sequences

- We could see the situation of continuity of a function firstly and intuitively by SEQUENCES then give a proof using limit(sequential criteria of limit), squeeze Thm(directly or constructing sequences),

e.g Thomas function

e.g

My main Analysis learning instructions(philosophy): Theorems come, theorems go, only examples lie forever.

⟵ **Proof idea** Identify the exact statement before proving it.

Maybe I could not remember all process of each profound theorems proof, but I could also understand Analysis in depth by handling how to use theorems to solve problems.(But I need to try to know how the proof using each condition in the theorems and internalize it by continuousely thinking) In this case, examples help me a lot.

数学素养和人文素养的结合一个好的证明是 natural 的好的数学是具有普遍性 (universal) 的（大数学家先看二维再推广到 n 维）sequence is the most 直观来写分析北极是无穷大上帝是无穷唯一的上帝正无穷和负无穷最后归到一点两个无穷大傅立叶级数周期函数–周期概念傅立叶 transform

偏导数存在不一定连续只有在一元才是对的

open set is countable union of open intervals—

欧拉拉格朗日比较有创意，数学的趣味，但后面，cauchy weiastrass 给的连续定理，严格化

记得写出存在,（ps：存在主义是生命价值驱动的好心情源泉）

⟵ **My thought** The theorem is easier to remember when its hypothesis feels necessary.

‘Existence’ means beyond any human efforts, and objectively existence. We take the value that we need means a human effort, and because it is arbitrary, we have the freedom to choose one. — Dr. Zhang

max and min need to be found

eg. Finite open sets’ intersection is also open because we can choose the minimum ϵ

open set is like interval

Theorem

Let $O \subset R$ be an open set then $O = \cup_{n=1}^{\infty} I_n$ where I_n is an open interval in R

Def

A collection of open sets covers a set A if $A \subset \cup O_\alpha$. The collection $\{O_\alpha\}$ is called an open cover of A.

Theorem

Let C={O_α} be a collection of open sets of real numbers then there is a countable subcollection {O_i} of C such that $\cup_{O\in C}O = \cup_{n=1}^{\infty}O_i$

Any open cover of a set of real numbers contains a countable subcover.

⟵ **Motivation** Why does this definition matter later? Try an example and a counterexample.

i have proved that:

the closure of Q = R <-> $R \approx Q$ <-> $\forall x \in R, \exists x_n \in Q$ such that$x_n --> x$. Q is a countable, dense subset of R, R is separable.

accumulation point is different to limit point

eg. $a_n = (-1)^n(n+1)/n$, 1 and -1 are accumulation points but not limit points because the sequence is not convergent

the usage of proof by contradiction in the proof of limit

eg.

⟵ **Proof idea** Begin with epsilon-delta or a sequence characterization.

albel: guass is a fox, 走过的地方狐狸尾巴扫掉了

algebra of continuity

$\exists\delta = min\{\delta_1, \delta_2\}$

Continuity

history

cauchy 1821(decrease indefinitely with those of α)–weierstrass 1874 ($\epsilon - \delta$)—–

understanding in a long process

- δ decides how good the continuity of the function is.

Preface before

Cauchy 1822

What is a limit?–number

Until 1870, the answer to “What is the number” had been known.

(这样来看，学习数分的一年痛苦就不算什么)

实数是什么

实数是一个唯一的一个 complete ordered field 如果有别的它一定和实数是等价的

- dedekind 思想推广到 n 维空间有一定难度（limit inf；limit sup 这种大小只有 1 维空间才有）所以我们有了 cantor 的思想是更高一级的可以推广到更 general 空间 (norm space;metric space 所有 metric space 的证明思想和构造时候一模一样方法固定如此)

⟵ **My thought** The theorem is easier to remember when its hypothesis feels necessary.

order-cardinality

1，2，3，n 2，4，6，2n

f(n)=2n

- explaination：无穷大加一个数还是等于无穷大

$f:N->N\cup\{a_1,a_2,a_3\}$

f(1)=a_1 f(2)=a_2,..f(4)=1, f(n+3)=n, this function is one-to-one – bijection

–无穷大加上任意一个实属 x 还是无穷大

N x N is countable

$S\subset N$ S is countable

f(m,n)–>2^m3^n（只要取互质的两个数就 ok）

N x N–> N

f: 1-1

$2^{m_1}3^{n_1}$=$2^{m_2}3^{n_2}$—$m_1=m_2,n_1=n_2$

f(N x N)⊂ N

只要映射到 N 的 subset 就可以了

2 维 3 维 n 维

(0,1) is uncountable 反证法

Cantor’s diagonal process

欧几里得是第一个反证法—有无穷多个质数

{p1 p2 p3...}

p=p1+p2+pn+...+1

p1|p...pn|p—p is a prime—contradiction to countable

(0,1)={x1,...x} countable

x1=01a11a12a13. a1n∈{0,1,2,...9} x2=01a21a22a23. a2n∈{0,1,2,...9}

construct a real number

y=0.b1b2b3...

b1 budengyu a11 y budengyu x1

b2 budengyu a22 y budengyu x2

b3 budengyu ann y budengyu xn

bn in {a1,...a}–> y∈(0,1) but y∉$\{x_1,...x_n\}$

Ex: [0,1) 约等于 (0,1)

proof: Let{p1,p2,....}=(0,1)∩Q 分成有理数和无理数

f(0)=p1 f(p1)=p2 f(pn+1)=pn f(x)=x $x\in(0,1)-${p1,..pn}

概率特征函数

Cantor set

进位的概念

导数积分无穷级数 –都是极限极限是什么–数数是什么 – Augustin Cauchy(1789-1846)

- 人文素养

cantor 1872 年是不平凡的一年，有（Weierstrass, Dedekind, Meray, Heine, Cantor）同时提出了实属系统建构的理解出版著作解决困惑人们 2500 多年的问题

Weierstrass 法律数学喝酒干架退学在中学里边仍然没有与学术脱节复变函数椭圆函数（研究在复数平面上的解析函数，C=R^2 a+ib–>(a,b) is an isomorphism）$C \cup \infty$ 相似于球面 S^2 约等于 $R^2 \cup \infty$ 变成 closed and bounded one-point- compactfication 为什么实数不 bounded 因为无穷无穷大这点不是实数（不满足实数性质—无穷大加无穷大还是无穷大；无穷大减无穷大是 any number）因为

任何单调有界的实属序列都有极限存在—dedekind 的假设

dedekind cut

if rational number

毕达哥拉斯的年代的万物皆数是有理数

古希腊哲学古希腊数学

Thales

做测量工作创出非欧几何这门学问真正的天才是能够创出一门学问的

牛顿是科学第一人当人们迷迷糊糊的时候他打开理性的光

这一切是黑暗的上帝说让牛顿去吧于是就有了光

改编圣经的话–文采

毕达哥拉斯定理的证明都是非常有创意的东西

毕达哥拉斯定理十个证明

读慢一点

心浮气躁

数学是需要耐心的

一个城市的发展不应该是盖房子应该是好的博物馆音乐厅公园提高人民素养

volume 2 多变数 stolze 定理 divergent 定理

limit sup limit inf –这个东西就是我的实数

Cantor-== 柯西序列。距离概念

$|a_m - a_n| - - > 0$. d(a_m, a_n–>0) metric

equivalent equation equivalent class 实数实数是一个集合

cantor–集合论

定义是最难的

有共通之处–作为定义

会用定理很重要不是高斯

用多了就知道是怎么一回事

定理给了条件哪里用到什么条件怎么用的

碰到好老师的重要性姜立夫量纲自然

complete 的完备不是完美不是门掉了一个螺丝找一个螺丝完完全全放上去发挥作用

认识自己我是螺丝完完全全发挥我的作用把长处恩赐发挥出来就是 complete

有理数在实数中有很多孔隙把空隙。把 accumulation point 收紧来

closure of A = B

$\forall b \in B, \exists \{a_n\} \subset A$ such that $a_n - - > b$

$A∩B $ not equals to $□∀□> 0 $

实数是唯一一个 complete ordered field

代数结构—分配律

eg.complex number 没有 order 所以我们定义 norm

Complex numbers lack an order compatible with field operations, so their norm supplies a useful notion of size.

和闭区间套定理极限思想的联系和区别???

Thanks

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

- Thanks to Professor Zhang for teaching us MTH117.
- Thanks to Professor A Kun for giving our XJTLU students chances to learn from him, especially his sophisticated personal understanding of Mathematics(and his useful and effective exercise of Analysis). # sup 之前的好题

If A is an infinite subset of N, then |A| = |N|,i.e. A is countably infinite.

An infinite subset of the natural numbers can be listed in increasing order, establishing countable infinity.

in fact, N could be any countable set

整理证明所有可以证明 sup 的方法

- 定理，对于所有 epsilon，epsilon，一般是取个几分之一夹在两者间
- 为了证明所有 upper bound 都大于等于某个值（sup），我们假设所有 upper bound 都小于等于这个值，然后推出矛盾（一般运用 Archimedean property，。。。）
- 假设存在 epsilon 大于 0，使 sup- epsilon 是 upper bound，然后根据题目条件，推出 sup-epsilon 不是 upper bound 的矛盾，（特别是某些集合中的特殊的值），得出 sup 是最小的 upper bound
- claim s=supS，然后把 s 的所有范围讨论（通过得到某些 n 使得 sups 不等于 s 的矛盾来排除 <,> 得到等号）

The supremum proof tests values below and above the proposed least upper bound to rule both out.

(Though using ϵ is the most quickly way but I do not like it at all because it does not show the essence or nature of the supremum or infmum, which also has the part of luck because maybe I understand it totally but because of my poor calculation skill about construction I will not solve it, leading to unhappiness to myself.)

总结了有上界（下界）必定有唯一上（下）确界的两种证明方法，闭区间套定理的证明，以及有限覆盖定理的证明（待 internalized）

⟵ **Proof idea**
Identify the exact statement before proving it.

keep going！## 用 sup 的唯一性证明根号只有唯一正解将 = 改为小于放进 set 里，通过 sup 的唯一性证明，sup（是 set 范围内用小于通过 limit 思想过渡到等于来求解这个极限值）

然后用 method4:claim s=supS，然后把 s 的所有范围讨论（通过得到某些 n 使得 sups 不等于 s 的矛盾来排除 <,> 得到等号）

Nested intervals give a construction of the supremum from successively tighter bounds.

The upper-bound construction narrows the possible location of a unique supremum.

(notice! The rational number set The rational numbers below the square root of two show why the rational field is not complete. does not satisfy completeness since if we choose a subset of Q, which is all rational numbers less than $\sqrt{2}$, when we restrict this region on Q, its supremum does not exist.)

if a not empty set has upper bound, then it must have the unique sup.

Prove: if a not empty set has upper bound, then it must have the unique sup.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

Method 1: (Completeness axiom)

$$X \neq \phi, \quad Y = \{y \in \mathbb{R} \mid \forall x \in X(x \leq y)\} \neq \phi$$

$$\forall x \in X, \forall y \in Y, x \leq y$$

$$\Rightarrow \exists c \in \mathbb{R}, \forall x \in X, \forall y \in Y,$$

$$x \leqslant c \leqslant y(\text{ Completeness axiom})$$

$$\Rightarrow (c \in Y) \wedge (\{\forall y \in Y \mid y \geqslant c\})$$

$$\Rightarrow c = \min Y$$

So c is the only sup. (uniqueness of minimum element of a set—2 inequalities lead to the equality leading to the only one result).

Method 2: Cantor Nested Interval Property(limit 思想).asdfaskfkasdjklsad

⟵ **Motivation**
What does the definition really control? Try to draw it before proving it.

We choose a random upper bound $\gamma, x \in E($ the set)

$$[x, r] = [a_1 b_1] \supset [a_2, b_2] \supset \cdots [a_n, b_n]$$

(each time we use the method of bisection to choose one side Including the point in E) $$

Each nested interval meets the set, so its limiting endpoint approaches the set supremum.

If a not empty set has lower bound, then it must have the unique inf.

Method 1: same as before.

Method 2: Based on before.

Suppose We choose m as a lower bound of E.

$$\begin{aligned}
&\Rightarrow \forall x \in E, x \geqslant m, -x \leqslant -m. \\
&\text{Let } F = \{-x \mid x \in E\}. \\
&\Rightarrow \beta = \sup F \\
&\Rightarrow -x \leqslant \beta, x \geqslant -\beta. \\
&\forall \varepsilon > 0, \exists -d \in E, -d > \beta - \varepsilon, \\
&\Rightarrow d < -\beta + \varepsilon \\
&\Rightarrow -\beta = \inf E
\end{aligned}$$

$$\Rightarrow -\sup(-E) = \inf E.$$

Method 3 more generalized than Method 2's conclusion (if c is negative then inf(cA)=csupA)

⟵ **Proof idea**
Identify the exact statement before proving it.

In fact, If $c < 0$, then $\sup(cA) = c\inf A, \inf(cA) = c\sup A$. (and **in particular. sup** $(-B) = -\inf B$)

Since if M=supA, $\forall x \in A, x \leq M, cx \geq cM$, which indicates that cM is the lower bound of cX.

So, cA has lower bound (s)if and only if A has upper bound(s).(inverse, also true) cA is not empty if and only if A is not empty.

So, cA has inf(cA) if and only if A has supA

$$\begin{aligned}&\forall cx \in CA, cx \geqslant cM\\ &\text{if } \exists cM', \forall cx \in cA, cx \geqslant cM', cM' > cM,\\ &X \leq M', M' < M(\forall x \in A)\end{aligned}$$

Contradicting to the condition that $M = \sup A$

So $\nexists CM'$ So cM is the largest lower bound

$$\begin{aligned}&\text{so } cM = \inf(cA)\\ &\text{so } c\sup A = \inf(cA)\end{aligned}$$

exercise: (1,2]'s sup–

Method 1

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

- 2 is an upper bound of [1,2) (obviously)
- if $\exists \varepsilon > 0$, s.t $2-\varepsilon$ is also an upper bound of $[1,2)$

$$\begin{aligned}&\because 1 \in [1,2)\\ &\therefore 2-\varepsilon \geqslant 1\end{aligned}$$

choose $2-\frac{\varepsilon}{2} \in (2-\varepsilon, 2)$ Then $2-\frac{\varepsilon}{2} \in [1,2)$ So $\exists\left(2-\frac{\varepsilon}{2}\right) \in [1,2)$ while $\left(2-\frac{\varepsilon}{2}\right) > (2-\varepsilon)$ So $2-\varepsilon$ is not an upper bound, contradicting to the suppose. So we have proved that 2 is the smallest upper bound, i.e. $\sup[1,2) = 2$.

Method 2

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

- 2 is an upper bound of [1,2) (obviously)
- $\forall \varepsilon > 0, \exists b \in [1,2)$ with $b > 2-\varepsilon$. We can take $b = \max\left\{2-\frac{\varepsilon}{2}, 1\right\}$.

exercise: supC=sup(A+B)=supA+supB from Professor A Kun

$$\begin{aligned}&\text{if } A \subset R, B \subset R, \text{ define:}\\ &C := A+B = \{z \in R : z = x+y, x \in A, y \in B\}\\ &D := A-B = \{z \in R : z = x-y, x \in A, y \in B\}\end{aligned}$$

show that

$$\begin{aligned}&(1)\ \sup C = \sup(A+B) = \sup A + \sup B\\ &(2)\ \sup D = \sup(A-B) = \sup A - \inf B\end{aligned}$$

(1) Proof:: Obviously,C has upper bounds if and only if A and B have upper bounds, and C is not empty. So C has sup C if and only if and only if A has sup A and B has sup B. (Completeness axiom).

- prove: $\sup C \leqslant \sin A + \sup B$.

$$\because x+y \leqslant \sup A + \sup B$$

$\therefore(\text{sip } A + \sup B)$ is an upper bound of C

$$\therefore \sin C \leqslant \sin A + \sup B$$

- Prove : $\sup C \geqslant$ supA $+$ sup B

$$\begin{aligned}&\because \forall \varepsilon > 0, \exists x \in A, y \in B, \text{ s.t.}\\ &\sup A - \varepsilon < x, \sup B - \varepsilon < y.\\ &\Rightarrow \sup A + \sup B - 2\varepsilon < x+y\\ &\text{i.e. } (\sup A + \sup B - 2\varepsilon)_{\max} < (x+y)_{\max}\end{aligned}$$

Since $x+y \leq \sup$ c We have $\sup A + \sup B - 2\varepsilon < \sup c, \forall \varepsilon > 0$

$$\text{i.e. } (\sup A + \sup B - 2\varepsilon)_{\max} < \sup C$$

We. As $\varepsilon \to 0$, We have $\sup A + \sup B \leqslant \sup C$ So, we have $\operatorname{Sup} C = \operatorname{Sup} A + \operatorname{Sup} B$. Then, we have $\sup D = \sup(A - B)$

$$\begin{aligned} &= \sup(A + (-B)) \\ &= \sup A + \sup(-B) \\ &= \sup A - \inf B \end{aligned}$$

(We have proved $\sup(-B) = -\inf B$ before)

another example

⟵ **Proof idea** Identify the exact statement before proving it.

$$a^* - \epsilon < a < a^* \quad \text{and} \quad b^* - \epsilon < b < b^*$$

Then

$$(a^* - \epsilon)(b^* - \epsilon) < ab \leqslant a^* b^*$$

or ignoring ϵ^2 term (If ε > 0, then ε, 3ε, ε², ε⁵, they all represent “any number greater than zero”, and ε’ also represents any number greater than zero, so they are equivalent, that is, we can say that they are equal to ε’.)

$$a^* b^* - (a+b)\epsilon = a^* b^* - \epsilon' < ab \leq c^*$$

This is true for all $\epsilon' > 0$, so

$$\sup A \sup B = a^* b^* \leq c^* = \sup C$$

Combining the above two inequalities, we can conclude that $\sup A \sup B = \sup C$. ## an exercise about think good Archimedean number from Professor Andrew Lin(A Kun)

13. Consider the set

$$A = \left\{ (-1)^n \left(1 - \frac{1}{n}\right) \middle| \ n \in \mathbb{Z}^+ \right\}.$$

- Show that 1 is an upper bound for A.
- Show that if d is an upper bound for A, then $d \geq 1$.
- Use (a) and (b) to show that $\sup A = 1$. [Solution]:
- We will show that for any $x \in A, x \leq 1$. Since $x \in A$, then $x = (-1)^n(1 - 1/n)$ for some $n \in \mathbb{Z}^+$. Since $\frac{1}{n} > 0$, then $1 - \frac{1}{n} < 1$. We argue our desired inequality in two cases. If n is even, then $x = (-1)^n(1 - 1/n) = 1 - 1/n < 1$. If n is odd, then $x = (-1)^n(1 - 1/n) = -1 + 1/n < 0 < 1$. In either case, $x \leq 1$ (in fact, $x < 1$) and 1 is an upper bound for A.
- Let d be an upper bound for A. Thus, $(-1)^n(1 - 1/n) \leq d$ for all $n \in \mathbb{Z}^+$. Assume, to the contrary that $d < 1$. Thus, $1 - d > 0$. By the Archimedean Property, there exists an $n \in \mathbb{Z}^+$such that $1 < (1-d)n$. Since $n > 0$, we can rewrite this as $\frac{1}{n} < 1 - d$, which is equivalent to $d < 1 - \frac{1}{n}$. If n is even, then $(-1)^n = 1$ and we have that

$$d < (-1)^n \left(1 - \frac{1}{n}\right) \in A$$

contradicting the fact that d is an upper bound. If n is odd, then consider instead $n+1$, which is even. Then, $(-1)^{n+1} = 1$ and

$$d < 1 - \frac{1}{n} < (-1)^{n+1} \left(1 - \frac{1}{n+1}\right) \in A$$

This again contradicts that d is an upper bound for A. Either way, we reach a contradiction and therefore conclude that $d \geq 1$.

- obviously

lower bound + inequaility —smallest upper bound—sup

another exercise about another episilon from Professor A Kun

14. Find the least upper bound for the following set and

$$A = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \cdots, \frac{n}{n+1}, \cdots \right\}$$

[Solution]: We note that every element of A is less than 1 since

$$\frac{n}{n+1} < 1, \quad n = 1, 2, 3, \cdots$$

We claim that the least upper bound is $1, \sup A = 1$. Assume that 1 is not the least upper bound. Then there is an $\epsilon > 0$ such that $1 - \epsilon$ is also an upper bound. However, we claim that there is a natural number n such that

$$1 - \epsilon < \frac{n}{n+1}$$

This inequality is equivalent to the following sequence of inequalities

$$1 - \frac{n}{n+1} < \epsilon \quad \Longleftrightarrow \quad \frac{1}{\epsilon} - 1 < n$$

Reversing the above sequence of inequalities shows that if $n > \frac{1}{\epsilon} - 1$, then $1 - \epsilon < \frac{n}{n+1}$ showing that $1 - \epsilon$ is not an upper bound for A.

(if n > 1/ ϵ-1, $\nexists \epsilon$ s.t. 1-ϵ is an upper bound, contradicting to our suppose)

This verifies our answer.

← **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

Prove Cantor Nested Interval Property

Prove:

- $\exists c \in$ all closed internals
  - i.e. if the non-empty closed intervals $I_1 \supset I_2 \supset I_3 \cdots, \exists c \in R$, s.t. $c \in I_i, \forall i \in \mathbb{N}$, $c \in \bigcap_{i=1}^{\infty} I_i$
- If the limit of the lengths of these intervals are 0 then the point is unique
  - i.e. if$|I_n| \to 0$, c is unique

$$\forall \varepsilon > 0, \exists I_n (|I_n| < \varepsilon) \Rightarrow c \text{ !}$$

← **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Proof:

- proof of 1

$I_1 = [a_1, b_1]$

claim $\forall I_n = [a_n, b_n], I_m = [a_m, b_m]$ i.e. $a_n \leq b_m$

(if $a_n > b_m$, then $a_m \leq b_m < a_n < b_n$, which means they are separate internals without any intersection.)

$$\begin{gathered} \text{Let } X = \{a_n\} \text{ (left endpoint set)} \\ \text{Let } Y = \{b_m\} \text{ (right endpoint set)} \\ \forall a_n \in X, \forall b_m \in Y, a_n \leq c \leq b_m \end{gathered}$$

$\Rightarrow \exists c \in \mathbb{R}$, s.t., $\forall a_n \in X \ \forall b_m \in Y, a_n \leqslant b_m$.(completeness axiom) We then let $m = n \Rightarrow c \in I_n$

- proof of 2

Based on "Any implication is equivalent to its contrapositive"

$$\begin{gathered} \text{if } \exists c_1 < c_2, \text{ s.t. } c_1, c_2 \in I_n, \\ \forall n \in \mathbb{R}, a_n \leqslant c_1 < c_2 \leqslant b_n, \\ \Rightarrow |I_n| = b_n - a_n \geqslant c_2 - c_1 \end{gathered}$$

We then choose $\varepsilon = \frac{c_2 - c_1}{2} > 0$

then there exists no I_n s.t. $|I_n| < \varepsilon$, since

$|I_n| \geqslant c_2 - c_1 \geqslant \frac{c_2 - c_1}{2}$

So if $\exists I_n s.t. |I_n| < \varepsilon$, c!

prove finite covering lemma

Prove: Finite Covering Lemma

⟵ **Proof idea**
Identify the exact statement before proving it.

$$\text{Def: } S := \{X\}, \text{ (} x \text{ is a set)}$$
$$Y \subset \bigcup_{X \in S} X$$

We say. S is a cover of Y.

$$\text{i.e. } \forall y \in Y, \exists X \in S, (y \in X)$$

Finite Covering Lemma:

If $I : [a, b]$,

$$I \subset \bigcup_{n \in I} U_n, U_n = (\alpha_n, \beta_n)$$

$\exists U_1, \cdots U_k$, s.t., $I \subset \bigcup_{i=1}^{k} U_i$

(Summray of Finite Covering Lemma: A closed internal can be covered by finite number of open internals)

proof:

suppose $I = [a, b]$ could not be coverd by finite number open intervals:

Then we use the method of bisection to separate I,s.t.

$I = I_1 \supset I_2 \supset I_3 \cdots I_n \supset \cdots$ (all In can not be covered

$|I_n| = \dfrac{b-a}{2^n} \to 0$

Gover)

$\Rightarrow \exists : C \in \bigcap_{i=1}^{\infty} I_i$ (Cantor Nested Interral Property)

$c_i \in [a_i, b_i]$

$\Rightarrow C \in I \subset \bigcup U_i$(based on the given condition of the proof problem)

$\Rightarrow \exists U = [\alpha, \beta], s.t. c \in U \; let \; \varepsilon = min[c - \alpha, \beta - c]$

$\Rightarrow I_n \subset U$

The finite-covering argument selects a subinterval inside one open set and derives a contradiction.

which indicates that at least this I_n is covered by U, contradicting to the suppose.

so the finite covering lemma is true.

lecture notes

The lecture proof emphasizes that every inference needs a stated reason.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

1

everything must be a reason during the process of proof. Patience is necessary!

limit

real number

history

Ancient Greece philosophy

Pythagoras, Archimede, Euclide,etc.

maths behave a discpline

study maths for its own sake! - number theory Q: rational number: a/b, a,b ∈ Z, b no 0

Greek believes: “Atoms”–Q are all numbers

- geometry (length of sig ∈ Q)

discovery: $\sqrt{2}$ in a square or ... is not rational—kill

— no understanding of real number, irrational infinity... no corresponding of number and geometry

parallel—square using move

but it is not a possible thing to turn a circle into a square—π is not a rational number —–paradox: zhinuo and wugui—-convergence of a sum (they do not understand infinity and real number)

calculus

Newton Leibniz makes phy a idependent discipline

Newton’s Law–Kepler Law solve ODE:$GMm/r^2 = F = ma = mr^{\cdot\cdot}$

Philosophiae Naturalis Principil Mathematica. every theorem and lemma are detailed proved

–no understanding of real number but a little understanding of limit and infinity

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

analysis 19 century

Cauchy cantor weierstrass hilbert

understand real number infinity limit

the foundation of maths

Chaper 1: Logic(grammar) v.s. Maths(literature)

we should solve out logic first to logic important in–programming(some derivations of maths)

statement

p

⟵ **Proof idea**
Identify the exact statement before proving it.

—An assertion that is either true or false but not both

e.g.

Negation of a statement p is a statement which means the opposite of P ~p—-read “not p”

Quantifiers all, every, each,no(none)—universal quantifiers

some,there exists, there is at least on, etc.—existential quantifiers

P: *some* a's are b's

~P: *all* a's are not b's/ no a's are b's

P: *some* a are not b

~p: *all* a are b

Truth table: give the truth values of related statements in all possible cases (relevant to boolean in computer science)

compound statement: combining several statements via logical operations

Conjunction: p^q. p and q

Disjunction: p v q–p or q

one of them is true, p v q is true, so we only need to decide if oe of them is true, if it is, the p v q is true

p V (~p) is always true

a statement that is always true is called a tautology

P10,1.4

Equivalence of statements: Two statements are **logically equivalent**, if they have the same truth values in all possible situations.

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

$A \equiv B$

'abstract non-sense'

Them(De Morgan's laws)

A,B

$\sim (A \wedge B) \equiv (\sim A) \vee (\sim B)$

$\sim (A \vee B) \equiv (\sim A) \wedge (\sim B)$

draw truth table to look at values

交的话（and），都 T 才 T；并的话（or），1T 则 T

Conditional:

If p(hypothesis/assumption/condition), then q(conclusion/consequence/result).

read: p implies q/assume p, then q.

key point: the implication is False when the rule is broken

A	B	$A \to B$
T	T	T
T	F	F
F	T	T
F	F	T

Def: p, q,p–>q

1)Converse: q—>p,i.e. if q, then p

2)Contrapositive: $(_{q)->(}$p), i.e. if not q, then not p

Prop: (p–>q)≡ (($_{q)–>(}$p))(a statement and its contrapositive are logically equivalent)

p	q	$\sim p$	$\sim q$	$p \to q$	$\sim q \to \sim p$	$(p \to q) \equiv (\sim q \to \sim p)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

(always remember: implication fails only when the rule brokens instead of.....(others))

Proof by contradiction: we want to prove p—>q, we prove ($_{q)—–>(}$p) instead

proof: Assume ~q, if , then if , then ... –> ~p, which is contradict to the original assumption p

e.g.: n is a natural number. Prove that if n^2 is divisible by 2(p), then n is divisible by 2(q).

Proof: Assume that n is not divisible by 2(~q),—> n is odd(defination), i.e. n =2k+1,k$\in Z$

—> n^2 =(2k+1)^2(multiplicaiton)=2()+1, which is odd

—> n^2 is not divisible by 2(~p),

Thus the aassumption that n is not divisible by 2 is false so n is divisible by 2.#

biconditional: p,q,

(p–>q)∧(q—>p)—p <–> q—-reads‘p if and only if q’.

statement converse

p	q	$p \to q$	$q \to p$	$p \leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Them.: the only case p<–>q is true is then both p and q ture or false

Set Theory p13

proposition

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

$\forall x, y \in \mathbb{Q}$, x is less than y, $\exists z \in Q$ s.t. x<z<y.

proof: $x, y \in Q, x < y$,x=m/n,x=p/q,z=x+y/2∈(x+x/2,y+y/2),i.e.x<z<y,z=pn+mq/2qn∈ $\mathbb{Q}$ by defination.

Russel paradox

let R be the set of all sets that are not a member of themself. i.e.R={S|S$\notin$S}

so if $R \in R$,R$\notin$ R

if $R \notin R$, R$\in R$

if $x \in A$,then $x \in B$ is the subset defination of A $\subseteq$ B.

if $A \subset B$, A is a proper subset of B. i.e. $A \subset B$ if $A \subseteq B$ and $\exists x \in B$, s.t. $x \notin$ A

$N \subset Z \subset Q \subset R$

Proposition

the empty set is a subset of any set

proof:

method1: (truth table)

method2:

⟵ **Proof idea**
Identify the exact statement before proving it.

Def

the complement of A is denoted by A^c

the intersection $A \cap B$

the union $A \cup B$

Countability and Bijections

Countable Set: A set is said to be countable if it is either finite or has the same size (cardinality) as the set of natural numbers $\mathbb{N}$. A set is countably infinite if there exists a bijection (a one-to-one correspondence) between that set and $\mathbb{N}$.

Uncountable Set: A set is uncountable if no such bijection exists, meaning its cardinality is strictly greater than that of $\mathbb{N}$.

Bijection: A bijection between two sets A and B is a function $f : A \to B$ that is both injective (one-to-one) and surjective (onto).

The Problem: Showing that $\mathcal{P}(\mathbb{N})$ is Uncountable

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

Cantor diagonalization constructs a subset of the natural numbers missing from any proposed enumeration.

method 1:

For each natural number n, if n is not in the subset f(n), we add n to the below A set; Otherwise, do not join. (just like the Cantor diagonal method (diagonal number is not in the range of the map so we add each of them into another set))

Consider the set $A = \{n \in \mathbb{N} \mid n \notin f(n)\}$, f : N → P(N).

we should prove it is not a bijection. so we could prove it is not surjective (P(N) is much bigger than N).

now we ask if f is surjective:

If yes, there exists $n_0 \in N$ s.t. f(n_0) =A, A⊂N while A ∈ P(N)

now we ask: Does $n_o \in f(n_0)$

1) if yes, $n_0 \notin A = f(n_0)$
2) if no, $n_0 \in A = f(n_0)$

(or we discuss the set, we suppose x is in A and based on the condition of being in set A we get A is empty, which is contradicts to "x is in A")

so f is not surjective, which means f could not be bijective. so p(N) is uncountable.

method2 (just back to method1's way of construction): thinking but not complete and maybe not true——suppose the power set is countable:(analogy to Cantor's diagnal method but we do not choose diagnal now)

for elements in the power set that has finitely many numbers, we repeat the last number to be infinity; for the infinity we just write down them directly—-all of them could correspond to a natural number unniquely by our assumption of countable property. (but this is meaningless for sovling this)

Then we could find a set that is not in the previous correspondence

- eg of a bijection between 2 sets(namely 2 sets with the same size) see my Youtube vedio:

A bijection pairs elements of two sets so that each has exactly one partner.

The companion example checks whether the proposed correspondence is one-to-one and onto.

an eg question-Logical Puzzle on Truth of Statements relavent to truth table

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Consider the following 99 statements:

- S_1: “Among these 99 statements, there is at most one true statement.”
- S_2: “Among these 99 statements, there are at most two true statements.”
- …
- S_{99}: “Among these 99 statements, there are at most 99 true statements.”

The goal is to determine which statements among these 99 are true.

(hint:Consider the chain of implications from statement S_n to S_{n+1}. Which statement implies which?)

sol: we fail the hypothesis of $S_{n+1} \Rightarrow S_n$, but because

S_n	S_{n+1}	$S_n \Rightarrow S_{n+1}$
T	T	T
T	F	F
F	T	T
F	F	T

Thus, we conclude that the truth of S_n implies the truth of all subsequent statements S_{n+1}.

)

$$S_n \Rightarrow S_{n+1}$$

so there exists an S_n, after which are true while before which are false.

so:There are $100 - n$ true statements.

$$100 - n \leq n \implies n \geq 50$$

There are $n - 1$ false statements.

$$n - 1 + 1 \leq 100 - n \implies n \leq 50$$

so The 50th statement S_{50} is true. Statements S_1 to S_{49} are false, and statements S_{50} to S_{99} are true.

Cardinality

⟵ **Proof idea**
Identify the exact statement before proving it.

- numebr of elements in a finite set: Def: let A be a set, if A contains finitely many elements, then the number of elements of A is called the cardinality of A, denoted by |A|(|A|$\in Z_{\geq 0}$
- A, B have infinitely many elements. If $\exists f : A- > B$ is a bijective map, then we regard the A, B have the same “cardinality”, i.e.|A|=|B|

A x B-Cartesian product

eg

|A|=3, |B|=2, |A x B|=6

map

Question from the Cardinality:

What happens if A has infinitely many elements?

Def: LEt A and B be 2 sets, A map(function)f: A–>B assigns each element in A

there has 3 kinds of maps(see 107)

eg

the graph of f(x) in A x B is f: A–> B(x $\in A$,y$\in B$)

inverse defination

if f is a bijective map, so as f^{-1}.

countable

- A set A is called countable if A iseither a finite set of there exists a bijective map f: A to N.(Otherwise is uncountable)

eg

- $B = [x|x = 2n, n \in N$ is countable

proof: for x|–>x/2 of B–>N i) injective , $∀x_1 $ not = x_2, f(x_1)=x/2, = f(x_2)

- Z is countable (负无穷到正无穷范围内的整数)

f:N–>Z(n|–>n/2 if even, 1-n/2 if odd)

n|–>n/2 if even makes the positive integer possible

1-n/2 if odd makes negative integer possible

and both of them are bijections

so Z is countable

the real number

Limit(Topology version)

triangle inequality is a strong tool

$$\lim_{x\to\infty}\left|\frac{\frac{11}{x}-\frac{9}{x^2}}{25+\frac{35}{x}+\frac{6}{x^2}}\right|$$

$$|25+\frac{35}{x}+\frac{10}{x^2}|=\left|25-\left(\frac{35}{-x}-\frac{10}{x^2}\right)\right|$$

$$\geqslant |25|-|\frac{35}{x}+\frac{10}{x^2}|$$

$$\geqslant 25-\left|\frac{35}{x}\right|-\left|\frac{10}{x^2}\right|$$

$$\forall x > 10$$

$$25-|\frac{35}{x}|-\left|\frac{10}{x^2}\right| > 25-\left(\frac{35}{10}+\frac{10}{10^2}\right)=M_1$$

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Take $M_2 = \frac{22}{M_1\varepsilon}, \forall|x| > \frac{22}{M_1\varepsilon}, \left|\frac{11}{x}\right| < \frac{M_1\varepsilon}{2}$ Take $M_3 = \sqrt{\frac{18}{m_1\varepsilon}}, \forall|x| > \sqrt{\frac{18}{m_1\varepsilon}}, \left|\frac{9}{x^2}\right| < \frac{m_1\varepsilon}{2}$

$$\left|\frac{11}{x} - \frac{9}{x^2}\right| < \left|\frac{11}{x}\right| + \left|\frac{9}{x^2}\right|$$

So Take $M = m \cdot ax\left\{10, \frac{22}{M_1 a}, \sqrt{\frac{18}{m_1\varepsilon}}\right\}$, we hame $\forall\varepsilon > 0, \forall|x| > m$, such that...<$\epsilon$

understanding deeply(explain by myself...)

proof of some theroem

⟵ **Proof idea**
Identify the exact statement before proving it.

boring things(just practivce epsilon-delta language...)

- $\lim_{x\to a} f(x) = \ell$ if and only if $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = \ell$.

Necessary Condition Let $f(x) \to l$ as $x \to c$.
Then from the definition of the limit of a function:

$$\forall\epsilon > 0 : \exists\delta > 0 : 0 < |x - c| < \delta \Longrightarrow |f(x) - l| < \epsilon$$

So for any given ϵ, there exists a δ such that:

$$0 < |x - c| < \delta$$

implies that:

$$l - \epsilon < f(x) < l + \epsilon$$

Now:

$$\begin{array}{ll} & 0 < |x - c| < \delta \\ \rightsquigarrow & -\delta < -(x - c) < 0 \\ & \vee \quad 0 < (x - c) < \delta \\ \rightsquigarrow & c - \delta < x < c \\ & \vee \quad c < x < c + \delta \end{array}$$

That is: $\forall\epsilon > 0 : \exists\delta > 0 :$
(1): $c - \delta < x < c \Longrightarrow \|f(x) - l\| < \epsilon$
(2) : $c < x < c + \delta \Longrightarrow \|f(x) - l\| < \epsilon$
So given that particular value of ϵ, we can find a value of δ such that the conditions for both:
(1): f tending to the limit l as x tends to c from the left
and :
(2) : f tending to the limit l as x tends to c from the right.
Thus:

$$\lim_{x\to c} f(x) = l$$

implies that:

$$\lim_{x\to c^-} f(x) = l$$

and:

$$\lim_{x\to c^+} f(x) = l$$

Sufficient Condition
Let $f(x) \to l$ as $x \to c^-$and $f(x) \to l$ as $x \to c^+$.
This means that:
(1): $\forall\epsilon > 0 : \exists\delta > 0 : c - \delta < x < c \Longrightarrow |f(x) - l| < \epsilon$

and :

(2): $\forall \epsilon > 0 : \exists \delta > 0 : c < x < c + \delta \Longrightarrow |f(x) - l| < \epsilon$

In the same manner as above, the conditions on δ give us that:

$$\begin{array}{ll} & c - \delta < x < c \\ \wedge & c < x < c + \delta \\ & 0 < |x - c| < \delta \end{array}$$

So:

$$\forall \epsilon > 0 : \exists \delta > 0 : 0 < |x - c| < \delta \Longrightarrow |f(x) - l| < \epsilon$$

Thus:

$$\lim_{x \to c^-} f(x) = l$$

and:

$$\lim_{x \to c^+} f(x) = l$$

together imply that:

$$\lim_{x \to c} f(x) = l$$

Cantor Set

decimal scale

$x = 0.a_1 a_2 a_3 ...$

[0,1]=$\{\sum_{n=1}^{\infty} a_n / 10^n, n \in 0, 1, ...9\}$

[0,1]=10^{∞}—(uncountable)

$|\{x_1\}| \subseteq (x_1 - \epsilon, x_1 + \epsilon), \forall \epsilon > 0$ i.e.$|\{x_1\}|$=0

⟵ **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

What is compactness?

Fresch <-> H.Lebesgal

Defination

Topology

$|x_n - x|$->0

$||x_n - x||$ “norm”

norm–the generalization of the absolute value: ||·||: V(a vector space)–>[0,∞)

(1) ||$\vec{x}$$≥$0||, $\forall \vec{x} \in V$

$||\vec{x} = 0||$ iff$\vec{x} = 0 \in V$

(2) $||\alpha \vec{x} = |\alpha| ||\vec{x}||$ $\alpha \in \mathbb{R}$

triangle inequality

水一个 commit

Here, be careful to choose the min delta

In a continuity argument, choose the smaller of two candidate delta values so both requirements hold.

The second continuity example again uses a common delta that satisfies every local bound.

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

连续性 + 用 sup 构造数列

A sequence chosen near a supremum approaches it while continuity transfers the limiting value.

用 sequential criterion 证明连续性取合适的 epsilon 来说明数列存在

eg.

A={1/2n} where n $\in N$

f(x)=0 when x $\in [0,1]$ and x is not in A

f(x)=2x when x $\in A$

(1) Prove f is continuous at x when x $\in [0,1]$ and x is not in A

- notice!!! Do not choose sequence convergent to a point that we want blindly, since there may exists no such sequence like that! For example, here, by the density of rational/irrational number, there exists no sequence composed by elements in A convergent to a point at [0,1]\A so we should think of another way to prove this—If b$\in [0,1] \setminus A$, then there is an n_0 such that$ 1/2(n_0+1)<b<1/2n_0,$ i.e. b$\in (1/2(n_0+1), 1/2n_0)$. For any sequence $a_n ∈[0,1] $with $ lim_n a_n=b,$ there is an N>0 such that $a_n \in (1/2(n_0+1), 1/2n_0)$. (It works as the following: We take an $\epsilon = 1/2 min|1/2(n_0+1)-b|, |1/2n_0 - b|, then there exists an N.0 such that |a_n - b| < \epsilon$. ===> when n>N, a_n$\in (b-\epsilon, b+\epsilon) \subset (1/2(n_0+1), 1/2n_0))$ Notice that (1/2(n_0+1),1/2n_0) does not intersect with A. Therefore f(a_n)=0. lim_nf(a_n)=0=f(b). Hence f is continuous at b.
- method 1 sequential
- method 2

(2) Prove f is not continuous at each point on A

- method 1 limit or $\epsilon - \delta$
- method 2 sequential creterion

Though below are equivalent:

α is an upper bound of f and α is a a functional limit of one or some sequences in the domain of f —-α is the supremum of f.

⟵ **Proof idea** Identify the exact statement before proving it.

Sup is different from limit – an experience from final exam

The n in them are different—

$\tan^{-1} x \sin x^2$

its supremum exists, which is $\pi/2$, however, its limit does not exist.

Reason: for sup we only need one "existence of x_0" since it says $\forall n \in N \; \exists x_0 \in R$ s.t. $M < f(x_0) < M + \epsilon$

but for the limit we need all x greater than a number that $|f(x) - M| < \epsilon$

For this specific example, we know that each top of $\sin x^2$ we have satisfied $|f(x) - M| < \epsilon$ but NOT ALL!

eg question from Dr. Chi-Kwong Fok about Weirestrass-Extream Value THM (continuous function in a compact set attains its max and min)

If f is a continuous and non-surjective function and the supremum of it in [0,+∞] is not in its range. show that $∀□>0, ∀x > 0, ∃z ∈R with z > x, ∀$

Firstly, negate it: $\exists \epsilon_0 > 0$, s.t. $\exists x_0 > 0, \forall z \in R$ with z>x_0, f(z) $≤M-□_0, $ where M is sup R(f) —(1)

Secondly, write down the def of M is sup:

⟵ **My thought** Which quantifier comes first? The order of choices often decides the theorem.

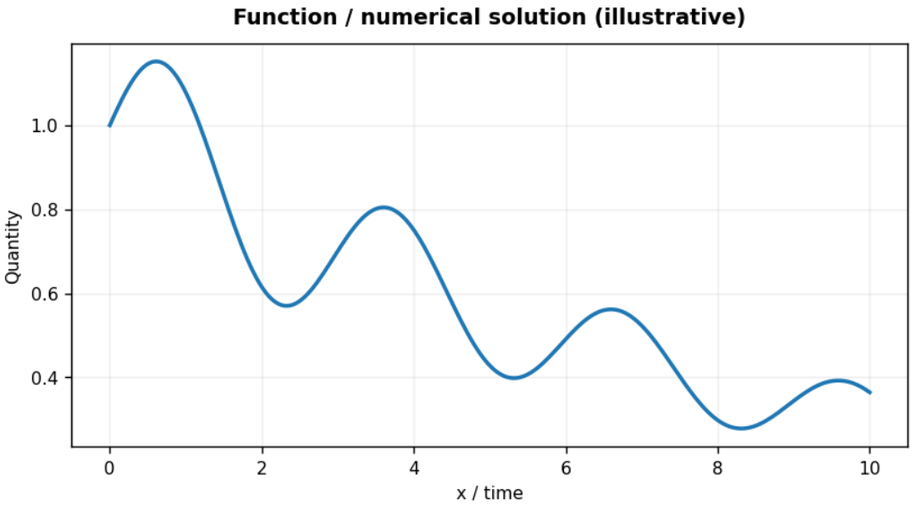


Figure 2: Output from code example 2. Illustrative reconstruction based on nearby code; not the exact original output. R packages: base R; functions: plot.

$\forall \epsilon > 0, \exists x \in [0, +\infty]$, s.t. f(x)>$M - \epsilon$. Now take the global $□$ to be ϵ_0 —(2)

By (1) and (2) we know that $\forall \epsilon > 0, \exists x \in [0, x_0]$, s.t. f(x)>$M - \epsilon$ Now take the global $□$ to be ϵ_0

By Weirestrass-Extream Value THM (continuous function in a compact set attains its max and min) we know it attains its maximum in [0,x_0]. Suppose this maximum is not the sup, then M=supR(f) < its maximum, then by the IVP it must attain sup R(f) contradiction. =–also. —contradiction

a problem about divisible using greatest common factor to solve(even Bezou's theorem)

⟵ **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Q:"Proof that if a positive integer p is not a perfect square, then $\sqrt{p}$ is irrational."

Sol: Proof by Contradiction:

Assume $\sqrt{p}$ is rational.

Since p is not a perfect square, there exist two coprime positive integers m and n with $n > 1$ such that

$$\sqrt{p} = \frac{m}{n}$$

Then

$$p = \frac{m^2}{n^2}$$

which implies

$$m^2 = n^2 p$$

i.e. $n^2 \mid m^2$. Then We want to show that $n \mid m$:

Suppose $m^2 = kn^2$, where $k \in \mathbb{Z}$.

Since m and n are both integers, based on prime factorization, we have:

$$m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

$$n = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$$

where p_i are prime numbers, and a_i, b_i are non-negative integers.

So,

$$m^2 = p_1^{2a_1} p_2^{2a_2} \cdots p_k^{2a_k}$$

and

$$n^2 = p_1^{2b_1} p_2^{2b_2} \cdots p_k^{2b_k}$$

Since $m^2 = kn^2$, we have:

$$p_1^{2a_1} p_2^{2a_2} \cdots p_k^{2a_k} = k \cdot p_1^{2b_1} p_2^{2b_2} \cdots p_k^{2b_k}$$

This implies:

$$2a_i \geq 2b_i \quad \text{for all } i$$

So,

$$a_i \geq b_i \quad \text{for all } i$$

Thus, m is divisible by n. Therefore, $n \mid m$, which is opposite to m and n are coprime. So, $\sqrt{p}$ is irrational.

method 2: Proof by Contradiction:

Assume $\sqrt{p}$ is rational.

Since p is not a perfect square, there exist two coprime positive integers m and n with $n > 1$ such that

$$\sqrt{p} = \frac{m}{n}$$

Then

$$p = \frac{m^2}{n^2}$$

which implies

$$m^2 = n^2 p$$

i.e. $n^2 \mid m^2$. So $n \mid m^2$

Since $n > 1$, it follows that there exists a prime number r such that $r \mid n$. (proof using Fundamental Theorem of Arithmetic:

Every integer greater than 1 is either a prime or can be uniquely factored into prime numbers.

If n is a prime number, then $r = n$ and clearly $r \mid n$;

If n is not a prime number, it must be decomposable into a product of prime factors. Therefore, we can write:

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$$

where p_i are prime numbers and e_i are positive integers.

Since n is a product of prime factors, at least one of these prime factors p_i must divide n.

Let $r = p_i$, which is one of the prime factors. Then $r \mid n$.

Thus, in both cases, whether n is a prime or not, there exists at least one prime number r such that $r \mid n$.)

Thus

$$r \mid m^2 \quad \text{and} \quad r \mid m$$

(proof of

$$r \mid m^2 \quad \text{and} \quad r \mid m$$

: proof1 using prime factorization as method 1's:

$r \mid m^2$ and r is a prime so $r = p_i$ corresponding to the exponential of $2a_i$

since $a_i \geq 0$, so a_i is at least 1 . proof2 using if r is a prime and r|ab, then r|a or r|b:

Suppose $r \nmid a$ and $r \nmid b$ (contradiction assumption).

Let $\gcd(a \cdot b, r) = d$ (where d is the greatest common divisor of $a \cdot b$ and r).

Since $d \mid r$, d is a divisor of r (since r is a prime number).

Therefore, d is either r or 1 (since r is prime).

if $d = r$, we have $\gcd(a \cdot b, r) = r$. This implies $r \mid a \cdot b$.

Since we assumed $r \nmid a$ and $r \nmid b$, it contradicts our initial statement based on Properties of greatest common divisor with its proof using Bezou's theorem;

If $d = 1$, we have $\gcd(a \cdot b, r) = 1$. This implies $r \nmid a \cdot b$, which contradicts our assumption that $r \mid a \cdot b$.

Thus, the contradiction shows that our assumption $r \nmid a$ and $r \nmid b$ must be false. Therefore, r must divide at least one of a or b.

)

Since m and n are coprime, this leads to a contradiction.

Therefore, $\sqrt{p}$ must be an irrational number.

everything goes smoothly in exams but not during the learning process

compact -"closed and bounded" is just appropriate for the real number set, but for the general metric space is not true—the real meaning is "every open cover has a finite subcover"

Gauss is a fox, and the fox's tail sweeps away the places he has passed through. —Abel

⟵ **Proof idea**
Identify the exact statement before proving it.

Gauss had the habit of not writing down his proofs, which made it difficult for others to understand his work. He had good habits of writing his daily work in his diary, which was a good way to keep track of his progress.

Read Euler, read Euler, he is the master of us all. —Laplace

Euler telled us both the true things and the false things in mathematics. He demonstrated the whole thinking process of mathematics.

Not only did Euler did good mathematics research but also he was a good teacher. He had cultivated lots of students. He wrote a lot of books and papers, which were very helpful for the development of mathematics.